



Complete Technical Setup & Architecture Manual

Zero-to-running walkthrough to build the Competitor Shift Radar & 10-Company Due Diligence Matrix on Windows.

1. System Prerequisites (One-Time Setup)

FREE SOFTWARE ONLY

1. Node.js (Version 18 or Newer)

Visit nodejs.org, download the LTS (Recommended For Most Users) Windows installer, run it, and accept the default settings.

2. Modern Web Browser & Notepad

You need Google Chrome, Microsoft Edge, or Brave to view the interactive matrix dashboard and Notepad to save your text files.

2. Creating Folders & Verifying Node.js

COMMAND PROMPT WALKTHROUGH

STEP A

Press **Windows Key + R**, type **cmd**, and hit **Enter**.

Open Command Prompt on Windows

STEP B

Copy and paste these exact lines into your Command Prompt window, then press Enter:

Create Project Folders on Drive E:

```
cd /d "E:\3sci\Work Hacks" mkdir "The Competitor Shift Radar" cd /d "E:\3sci\Work Hacks\The Competitor Shift Radar" mkdir data mkdir templates
```

STEP C

Copy your <https://lh3.googleusercontent.com/gg/ACRwjau3ghMV6H3rB1A4D7a83J4B4UZBQTD27kcyTGJqmSkehDFz-5cEe04biDv7011LfS9TLRwd8UDe0KaXdwUgDPvic0P5jc95E0Cp4g3S4KGanUMgWDMsuJva8tN0xw7ZA7TtwlCd5crFChq29CACQg8IyeiW3ij7wm5lGiYBAU4FGjUs5xTMTVnQ8M3aylcNg3mgttuVxZoJ4k3IsEP0PCWpt0sq1oj7ZT1PDAlJ9X5HY-rQjSV5MU50AnwIt1UWilHZ14wPURsn1AKhjc3H6Yt9dTHAPOH8hWg=s0> file into the newly created folder: **E:\3sci\Work Hacks\The Competitor Shift Radar\templates**

Place the Brand Logo

3. File #1: Package Manifest (`package.json`)

ROOT CONFIGURATION

Open Notepad, paste the exact JSON below, and click **File > Save As**. Set **Save as type** to **All Files (*.*)** and save as **E:\3sci\Work Hacks\The Competitor Shift Radar\package.json**:

```
{ "name": "the-competitor-shift-radar", "version": "1.0.0", "type": "module", "scripts": { "scan": "node build_dossier.js", "build:manual": "node build_manual_pdf.js" }, "dependencies": { "puppeteer": "^24.0.0" } }
```

STEP D

In your Command Prompt window, run the following command to download Puppeteer:

Install Background Engine Dependencies

```
npm install
```

4. File #2: Target Companies Roster (data/companies.json)

INGESTION SCHEMA

Open Notepad, paste this sample roster, and save it as `E:\3sci\Work Hacks\The Competitor Shift Radar\data/companies.json`:

```
[ { "company": "My Firm (Benchmark)", "is_benchmark": true, "products": "Rapid MVP Prototyping, AI Automation Consulting", "homepage": "https://example.com", "pricing_url": "https://example.com/pricing", "careers_url": "https://example.com/careers", "about_url": "https://example.com/about" }, { "company": "Target Alpha", "is_benchmark": false, "products": "Enterprise Workflow SaaS, API Integration Layer", "homepage": "https://alpha.com", "pricing_url": "https://alpha.com/pricing", "careers_url": "https://alpha.com/careers", "about_url": "https://alpha.com/about" } ]
```

5. File #3: Headless Web Extractor (build_dossier.js)

NODE.JS AUTOMATED SCRIPT

Open Notepad, paste the complete script below, and save it as `E:\3sci\Work Hacks\The Competitor Shift Radar/build_dossier.js`:

```
import fs from 'fs'; import path from 'path'; import puppeteer from 'puppeteer'; async function crawlPageText (page, url) { if (!url) return ''; try { await page.goto(url, { waitUntil: 'domcontentloaded', timeout: 25000 }); return await page.evaluate(() => { document.querySelectorAll('script, style, nav, footer, noscript, svg').forEach(el => el.remove()); return document.body.innerHTML.replace(/\s+/g, ' ').trim(); }); } catch (err) { return `[Crawl Warning: ${err.message}]`; } } async function runCompetitiveRadar() { const configPath = './data/companies.json'; if (!fs.existsSync(configPath)) { console.error('Error: ./data/companies.json not found!'); return; } const companies = JSON.parse(fs.readFileSync(configPath, 'utf-8')).slice(0, 10); console.log(`Loaded ${companies.length} targets for extraction.\n`); const browser = await puppeteer.launch({ headless: 'new', args: ['--no-sandbox'] }); const page = await browser.newPage(); await page.setUserAgent('Mozilla/5.0 (Windows NT 10.0; Win64; x64) Chrome/120.0.0.0'); const intelligenceRecords = []; for (let i = 0; i < companies.length; i++) { const c = companies[i]; console.log(`[${i + 1}/${companies.length}] Analyzing: ${c.company}`); const homeText = await crawlPageText(page, c.homepage); const pricingText = await crawlPageText(page, c.pricing_url); const careersText = await crawlPageText(page, c.careers_url); intelligenceRecords.push({ company: c.company, is_benchmark: !!c.is_benchmark, products: c.products || 'Not specified', summary: { homepage_snippet: homeText.slice(0, 600), pricing_snippet: pricingText.slice(0, 600), careers_snippet: careersText.slice(0, 600) } }); } await browser.close(); let matrixMd = `# 3Sci Multi-Company Due Diligence Matrix\n\n`; matrixMd += `**Scan Date:** ${new Date().toLocaleDateString()}\n\n`; matrixMd += `| Company | Role | Products / Services | Homepage | Pricing | Careers |\n\n`; matrixMd += `| :--- | :--- | :--- | :--- | :--- | :--- |\n\n`; for (const item of intelligenceRecords) { const role = item.is_benchmark ? '**Benchmark**' : 'Target'; matrixMd += `| **${item.company}** | ${role} | ${item.products} | ${item.summary.homepage_snippet.slice(0, 80)}... | ${item.summary.pricing_snippet.slice(0, 80)}... | ${item.summary.careers_snippet.slice(0, 80)}... |\n\n`; } fs.writeFileSync('competitive_matrix.md', matrixMd, 'utf-8'); console.log('\nAnalysis complete: competitive_matrix.md created successfully.');
```

6. File #4: Interactive App & AI Studio (`index.html`)

ZERO-BUILD HTML5 / CSS / JS

Open Notepad, paste the complete code structure below, and save as `E:\3sci\Work Hacks\The Competitor Shift Radar\index.html` :

```
<!DOCTYPE html><html lang="en"><head><meta charset="UTF-8"><title>Competitive Radar</title><link href="http
s://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@400;600;700;800&family=Space+Grotesk:wght@700&family=Je
tBrains+Mono:wght@400;600&display=swap" rel="stylesheet"><style>.root { --bg:#090f17; --card-bg:#111a26; --card-b
order:#1e2c3d; --accent:#00d2b4; --gemini:#4ade80; --text:#fff5f9; } * { box-sizing: border-box; margin:0; padding:
0; font-family: 'Plus Jakarta Sans', sans-serif; } body { background:var(--bg); color:var(--text); padding:25px; min-
height:100vh; } .grid { display:grid; grid-template-columns:460px 1fr; gap:20px; } .panel { background:var(--card-b
g); border:1px solid var(--card-border); border-radius:10px; padding:18px; } .card { background:#0d1520; border:1px
solid var(--card-border); border-radius:6px; padding:10px; margin-bottom:8px; } .card.is-benchmark { border-left:4p
x solid #38bdf8; } input, textarea, select { width:100%; background:#090f17; border:1px solid var(--card-border); c
olor:#fff; padding:6px 8px; font-size:12px; border-radius:4px; margin-bottom:6px; outline:none; } button { cursor:p
ointer; font-weight:700; border-radius:4px; border:none; padding:8px 12px; font-size:12px; } .btn-accent { backgrou
nd:var(--accent); color:#090f17; width:100%; } .btn-gemini { background:var(--gemini); color:#042f2e; } table { wid
th:100%; border-collapse:collapse; font-size:12px; } th, td { padding:8px; border:1px solid #1a2737; text-align:lef
t; } th { background:#0d1520; color:var(--accent); } .prompt-box { background:#090f17; border:1px solid #1a2737; pa
dding:10px; font-family:'JetBrains Mono', monospace; font-size:11px; white-space:pre-wrap; max-height:220px; overflo
w-y:auto; }</style></head><body><h1 style="font-family:'Space Grotesk', sans-serif; color:var(--accent); margin-bo
ttom:15px;">3Sci Competitive Radar & Prompt Studio</h1><div class="grid"><div class="panel"><h3 style="font-siz
e:13px; text-transform:uppercase; margin-bottom:10px;">Target Roster <span id="count">2</span>/10</h3><div id="c
ompany-list"></div><button style="background:#1c2b3d; color:#cbd5e1; width:100%; margin-bottom:6px;" onclick="addC
ompany()">+ Add Company (Max 10)</button><button class="btn-accent" onclick="renderAll()">⚡ Update Matrix & Promp
t</button></div><div class="panel"><div style="display:flex; justify-content:space-between; margin-bottom:10p
x;"><h3 style="font-size:13px; text-transform:uppercase;">Side-by-Side Comparison Matrix</h3><div><button onclick
="exportToWordDoc()">📄 Export .DOC</button><button onclick="window.print()">🖨 PDF</button></div></div><table id
="matrix-display"><thead><tr><th>Company</th><th>Role</th><th>Products / Services</th><th>Homepage</th><th>Pricing
</th></tr></thead><tbody id="tbody"></tbody></table><div style="margin-top:14px; background:#0d1624; border:1px so
lid #2e3e57; border-radius:8px; padding:12px;"><h4 style="color:#a78bfa; margin-bottom:6px;">⚡ Strategic Prompt S
tudio & Gemini 3.8 Flash Runner</h4><select id="persona" onchange="generatePrompt()"><option value="ma_auditor">R
uthless M&A Partner & Due Diligence Auditor</option><option value="headhunter">Executive Career Strategist & Hea
dhunter</option><option value="founder">Contrarian Founder & Competitive Moat Analyst</option></select><textarea id
="questions" placeholder="Your specific doubts (e.g. Can they undercut pricing?)..." oninput="generatePrompt()"></
textarea><div style="display:flex; gap:8px; margin:6px 0;"><input type="password" id="gemini-key" placeholder="Pa
ste free Google AI Studio key..." oninput="localStorage.setItem('3sci_key', this.value)"><button class="btn-gemin
i" onclick="runGemini()">⚡ Run Live AI</button></div><div id="gemini-output" style="display:none; background:#04
190e; padding:10px; border-radius:4px; font-size:12px; margin-bottom:8px; white-space:pre-wrap;"></div><div class
="prompt-box" id="prompt-display"></div></div></div><script>let companies = [ { name: "My Firm", is_benc
hmark: true, products: "Tech Studio, Rapid MVP Prototyping", homepage: "https://example.com", pricing: "https://exa
mple.com/pricing" }, { name: "Rival Alpha", is_benchmark: false, products: "Enterprise Workflow Automation SaaS", h
omepage: "https://alpha.com", pricing: "https://alpha.com/pricing" } ]; function renderCompanyList() { const contain
er = document.getElementById('company-list'); container.innerHTML = ''; document.getElementById('count').innerText =
companies.length; companies.forEach((c, idx) => { const div = document.createElement('div'); div.className = 'car
d' + (c.is_benchmark ? ' is-benchmark':'' ); div.innerHTML = `<input value="${c.name}" oninput="companies[${idx}].na
me=this.value" placeholder="Company Name"> <input value="${c.products}||'" oninput="companies[${idx}].products=thi
s.value" placeholder="Products / Services"> <input value="${c.homepage}" oninput="companies[${idx}].homepage=this.v
alue" placeholder="Homepage"> <input value="${c.pricing}" oninput="companies[${idx}].pricing=this.value" placeholde
r="Pricing URL"> <label style="font-size:11px;"><input type="radio" name="bm" ${c.is_benchmark?'checked':''}> onchan
ge="setBenchmark(${idx})"> Benchmark Firm</label>`; container.appendChild(div); }); } function setBenchmark(i) { co
mpanies.forEach((c, idx) => c.is_benchmark = (idx === i)); renderAll(); } function addCompany() { if(companies.leng
th < 10) { companies.push({ name: `Target ${companies.length+1}`, is_benchmark: false, products:"", homepage:"", p
ricing:"" }); renderAll(); } } function renderTable() { const tbody = document.getElementById('tbody'); tbody.inn
erHTML = ''; companies.forEach(c => { tbody.innerHTML += `<tr><td><b>${c.name}</b></td><td>${c.is_benchmark?'★ Benchm
ark': 'Target'}</td><td>${c.products}</td><td>${c.homepage}</td><td>${c.pricing}</td></tr>`; }); } function generate
Prompt() { const p = document.getElementById('persona').value; const q = document.getElementById('questions').valu
e; let txt = `Act as an expert (${p}). Analyze these companies:\n` + JSON.stringify(companies, null, 2); if(q) txt
+= `\nSpecific User Inquiries: "${q}"`; txt += `\nProvide: (1) Defensibility scorecard (1-10), (2) Pricing & white
space gaps, (3) 3 unasked leadership probes.`; document.getElementById('prompt-display').innerText = txt; return txt;
} async function runGemini() { const key = document.getElementById('gemini-key').value.trim() || localStorage.ge
tItem('3sci_key'); if(!key) return alert('Enter a valid Gemini API key. '); const out = document.getElementById('gem
ini-output'); out.style.display='block'; out.innerText='Analyzing with Gemini 3.8 Flash...'; try { const res = await
fetch('https://generativelanguage.googleapis.com/v1beta/models/gemini-3.8-flash:generateContent?key=${key}', { me
thod:'POST', headers:{'Content-Type':'application/json'}, body:JSON.stringify({ contents:[{ parts:[{ text:generateP
rompt() }] }] }); const d = await res.json(); out.innerHTML = d.candidates?.[0]?.content?.parts?.[0]?.text || d.
error?.message || `Error executing request.`; } catch(e) { out.innerHTML = `Network error: ` + e.message; } } funct
ion exportToWordDoc() { const blob = new Blob(['\uffeff', `<html><body><h1>Due Diligence Matrix</h1>${document.getEl
ementById('matrix-display').outerHTML}<h2>Strategic Analysis</h2><pre>${document.getElementById('gemini-output').in
nerText || document.getElementById('prompt-display').innerText}</pre></body></html>`], { type:'application/msword'
}); const a = document.createElement('a'); a.href = URL.createObjectURL(blob); a.download = `3Sci_Matrix_${Date.now
()} doc`; a.click(); } function renderAll() { renderCompanyList(); renderTable(); generatePrompt(); } window.onload
=runGemini();</script></body></html>
```



7. Step-by-Step System Verification

ZERO TO EXECUTION

STEP 1

Launch the Dashboard

Open Windows File Explorer, navigate to `E:\3sci\Work Hacks\The Competitor Shift Radar\`, and double-click `index.html`. It opens instantly in your default browser.

STEP 2

Test In-Browser Gemini Execution

Get a free key from aistudio.google.com/app/apikey, paste it into the password field, and click ⚡ **Run Live AI**. The analysis streams right into the page and stays stored in your browser.

STEP 3

Test Document Export

Click  **Export .DOC**. A formatted Word file downloads immediately containing the side-by-side matrix table and the live AI synthesis text.

8. Common Troubleshooting Solutions

QUICK RESOLUTION GUIDE

Error: "Cannot find package 'puppeteer'"

You ran the script before running installation. Run `npm install` inside `E:\3sci\Work Hacks\The Competitor Shift Radar\` to complete the setup.

Error: "Model gemini-2.5-flash not found"

Google retired older endpoints. Ensure your code targets `gemini-3.8-flash` in the REST URL. This is already implemented in File #4.

Browser shows raw code instead of page

Notepad accidentally saved the file as `index.html.txt`. Rename it in File Explorer to end strictly with `.html`.

API Key disappears on refresh

Make sure your browser is not in Incognito/Private mode, which clears `localStorage` automatically when windows close.

Security & Privacy Architecture

All web requests to the Gemini API occur directly from your browser to Google's official endpoint over encrypted TLS 1.3. No company targets, competitor URLs, private notes, or API keys are ever transmitted through or stored on 3Sci servers.

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